ReliableML Method Note: Model Deterioration over Time

Jabulani Monchwe, Olorato Motitswane, and Mfundo Monchwe

Version 0.1.1

Abstract

The deterioration method summarizes labeled prediction batches by accuracy, fits a linear accuracy trend over time, and bootstraps within-batch rows to estimate uncertainty in the slope.

1 Data and estimand

Each history record contains a timestamp, true labels, and predictions. At least three distinct dates are required. Timestamps must be consistently numeric or ISO date/time values. The method computes one accuracy observation per batch, sorts the observations by time, and fits a simple linear regression of accuracy on time. Each batch contributes one equally weighted point, regardless of its row count.

2 Bootstrap and flag

Within every batch, row indices are resampled with replacement and accuracy is recalculated. The linear slope is refit across dates for each bootstrap replicate. The report returns the observed slope, a percentile confidence interval, the regression p-value, and the number of bootstrap samples. A deterioration flag is raised only when the entire bootstrap slope interval is below zero.

3 Limitations

This is a descriptive temporal association, not a causal attribution to the model. It does not adjust for case-mix changes, operational changes, label delay, changing evaluation protocols, or serial dependence. The within-batch bootstrap assumes independent rows; clustered or temporal dependence can require block resampling. Use consistent, representative labeling and inspect both batch sizes and subgroup performance.

4 Interface

```
history = [
    {"timestamp": "2026-01-01", "y_true": y_jan, "y_pred": pred_jan},
    {"timestamp": "2026-02-01", "y_true": y_feb, "y_pred": pred_feb},
    {"timestamp": "2026-03-01", "y_true": y_mar, "y_pred": pred_mar},
]
report = model.assess_deterioration(history)
print(report.accuracy_slope, report.slope_confidence_interval)
```

5 Further reading

References

[1] J. Gama et al. A survey on concept drift adaptation. *ACM Computing Surveys*, 46(4), 2014. https://doi.org/10.1145/2523813.

[2] ReliableML deterioration protocol: https://github.com/Jabualani/ML-Reliabilty.